

# Changelog

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All notable changes to **Rev80** are documented here. Sections dated before the 2026-08 rebrand describe the project under its former name, vibechecker, and retain it as an accurate record. Format follows [Keep a Changelog](#).

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## [Unreleased] — feature/spectral-averaging (2026-08-29)

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### Added

- **Linear power averaging of the spectrum over N frames**, with an enable checkbox and N in the acquisition dialog (off by default). Welch's method *is* linear power averaging, but since the F-8 fix set `nperseg = blocksize` it runs exactly one segment per frame — so there was no averaging anywhere in the chain, and `welch_overlap` had nothing to act on. Each bin of a single-segment estimate is  $\chi^2(2)$ , with a standard deviation equal to its own mean, which is why the floor looks rough and a small line is hard to pick out of it.
  - `AcquisitionSettings.averaging_enabled / .n_averages / .n_averages_effective` (clamped to `cache_frames` — you cannot average more frames than are retained); `ChannelResult.n_averages` reports the count **actually achieved**.
  - A derived **Avg. Window** field reading `16 x 0.5 s = 8 s`, flagged when capped by the cache: N alone is hard to reason about, and how long the machine must stay steady is what the analyst actually needs to know. A frame is exactly `1/binsize` seconds.
  - A live “averaging 12 of 16 frames” line beside the peak count, showing “of N” only when the delivered count falls short.
  - `DataCollector._psd_and_overalls_for()` — the per-frame Welch PSD and 5-order overalls, extracted from `process_sample` so the averaging accumulator reaches earlier frames through exactly that code rather than a second copy. Two divergent copies of one path was the direct cause of M-05.
  - `tests/test_spectral_averaging.py` — 19 tests.

### The rule tying live and replay together

The average is the N most recent **valid** frames up to and including the frame being displayed. Live that is the last N received; browsing it is `frames[cursor-N+1 ... cursor]`. One rule, so stepping forward through a loaded file reproduces exactly what the live display showed at that moment — the same replay-fidelity property F-4 established for the high-pass.

**Nothing is baked into the file.** The HDF5 stores individual raw frames, so a capture taken with averaging *off* can be given 16 averages after loading, and changing N recomputes without reloading. Averaging is a view on stored data, not a property of it.

### Decisions, each pinned by a test

- **Power domain, never amplitude.** Average  $|X|^2$ , then sqrt at the end. On a coherent line the two are identical, which is exactly why this is easy to get wrong and stay green — it shows only on the noise floor, where averaging magnitudes converges about **11% low** (a Rayleigh magnitude has mean  $0.886 \cdot \sqrt{\mu}$  against the RMS). The overall combines as `sqrt(mean(squares))` for the same reason.
- **Above the per-frame PSD cache**, so changing N invalidates no per-frame work and does not have to join `psd_key`.

- **Overloaded records are rejected from the average**, including when the displayed frame is the overloaded one — analyzer practice, and the same reasoning F-5 used to keep them out of the trend. The waveform, the scalars and the overflow flag still come from the displayed frame, so nothing is hidden; only the spectral estimate is protected.
- **Crest factor and kurtosis are NOT averaged**. Averaging is for steady-state estimation; those exist to catch the frame that is *not* steady.

*Recorded because it is the point of the whole exercise: the first revert-check pass showed that swapping power averaging for amplitude averaging **passed all 17 tests**, despite the design note calling it the one thing that must be right. Two tests were added asserting against both candidate answers so the wrong one cannot pass.*

## Measured

Noise plus a 300 Hz line,  $F_{\max} 1000 / df 2$ :

| N  | floor CV | $1/\sqrt{N}$ predicted | floor level | line amplitude |
|----|----------|------------------------|-------------|----------------|
| 1  | 0.5260   | 0.5260                 | 0.03387     | 0.22366        |
| 4  | 0.2248   | 0.2630                 | 0.03626     | 0.22183        |
| 16 | 0.1189   | 0.1315                 | 0.03740     | 0.21965        |
| 64 | 0.0604   | 0.0657                 | 0.03800     | 0.21878        |

Scatter tracks  $1/\sqrt{N}$ ; the line is untouched at  $-2\%$  across a  $64\times$  change. The floor *level* rises 12% over that range — not drift, but the Rayleigh-mean-to-RMS convergence the power-domain argument predicts, and the clearest confirmation that the average is being taken correctly.

## Still open

`welch_overlap` remains inert: averaging across frames does not overlap segments *within* a frame. Making it real would roughly double the available averages for the same wall time.

## [Unreleased] — round 2: vibration-analysis integrity (2026-08-29)

Second remediation round from the three-discipline audit, addressing the vibration-engineering findings the first round did not cover. Round 1 fixed *how* the numbers were computed; this round fixes *what they were computed over*, and adds the diagnostics an instrument in this class needs.

## Added

- **envelope.py** — envelope (demodulation) analysis, the audit's only "blocks stated purpose" gap. `envelope_spectrum()` band-passes around a structural resonance, takes the Hilbert magnitude, removes the DC term and returns a coherent-gain-corrected amplitude spectrum; `suggest_band()` picks a demodulation band from the frame. A bearing defect's impulses ring a housing resonance at 2–20 kHz and are buried under the 1x in the raw spectrum, but appear as a clean line at the defect rate with  $\pm 1x$  load-zone sidebands in the envelope — months before the broadband overall moves. Verified end to end against the simulated oracle: auto band 2546–5046 Hz against a true 4000 Hz resonance, BPFO expected 325.8 Hz found at 324.0 Hz (within one bin) at **125x SNR**, with the 1x and lower sideband next. New **Envelope** plot tab with band controls and an Auto button.

- `_dsp.crest_factor()` / `_dsp.kurtosis()` and `ChannelResult.crest_factor` / `.kurtosis` — the two impulsiveness scalars a broadband overall averages away entirely. Non-excess kurtosis (Gaussian = 3.0), which is what condition-monitoring practice quotes. Computed on the band-limited displayed trace, never on the Hann-tapered array the overall uses, whose taper is an amplitude envelope that would corrupt any peak statistic. Trended, persisted to HDF5 and to the monitor session, and shown on each result card.
- **Declared measurement band** — `AcquisitionSettings.band_fmin` / `.band_fmax` with resolved properties, `util.ISO_BAND_PRESETS` (ISO 20816 10–1000 Hz and low-speed 2–1000 Hz), an acquisition-dialog preset combo plus editable edges, `ChannelResult.band_fmin` / `.band_fmax`, HDF5 and monitor-writer persistence, and the band shown on the result card and both trend axes.
- `simulation.GenerateBearingVibration()` — a physically realistic bearing-defect model: impulse train at the defect rate, each impulse ringing a structural resonance, amplitude-modulated at the shaft rate by the load zone, with cumulative slip jitter, and `severity=0` as the healthy negative control. Now the default `SimulatedSensor` source.
- `tests/test_declared_band.py` (46), `test_bearing_oracle.py` (20), `test_diagnostic_scalars.py` (13), `test_envelope.py` (14).

## Fixed

- **M-06 — the overall was not band-limited.** It was the RMS of the whole filtered block, so its band was `highpass_fc ... fs/2`, and `fs/2` is  $1.28\times\text{--}2.56\times$  maxfreq depending on where the power-of-two rounding in `samplerate` lands — **2.048× at the 500/1000/2000 Hz presets**. F-9 had already truncated the *spectrum* at maxfreq, so the number on the result card and the picture beside it described different bands, and nothing recorded which. Content the user had explicitly excluded via `F_max` still reached the trend (+25% on the audit's case, enough to cross an ISO 20816 zone boundary), and overalls were not comparable across sessions taken at different `F_max` — which silently invalidates long-horizon trending.
  - All five integration orders now share one masked, Hann-tapered path. Order 0 previously skipped the taper, correctly, because a passthrough performs no transform-domain multiply. Band-limiting removes that premise: the mask is such a multiply, hence a circular convolution in time.
  - The un-tapered Parseval alternative was measured and rejected: exact in band, but with a rectangular window's −13 dB sidelobes it let a 3× tone at 30 Hz past a 100 Hz edge at only **−22 dB**, inflating the overall +2.70%. Hann rejects it by **−84 dB**, and −100 to −144 dB elsewhere, for 4.9e-4 worst-case in-band error.
  - Hardware (4424A, AWG loopback, band 10–1000 Hz at `F_max` 2000): 800 Hz +0.0 dB, 950 Hz −0.0 dB, 1000 Hz −3.1 dB, **1100 Hz −58.2 dB**, 1300 Hz −58.6 dB, 1600 Hz −58.9 dB — a measured 58 dB cliff at the declared edge.
- **ISO 2954 — the high-pass −3 dB point sat on the declared band edge.** A 4th-order Butterworth designed at 10 Hz reads 29% low at 10 Hz, the very frequency the standard names as the bottom of its declared band. `highpass_fc` is now the declared edge and the knee sits below it at  $f_{\text{edge}} * (A^2 / (1-A^2))^{(-1/2N)}$ ; at  $N=4$ ,  $A=0.9$  that is  $0.8342 \times f_{\text{edge}}$ , so a 10 Hz edge designs at 8.34 Hz and reads −0.915 dB at 10 Hz. Safe only because the band mask now removes sub-band energy exactly, so the  $1/\omega^2$  blow-up the higher knee implicitly guarded against cannot reach the result. Hardware: 100 Hz +0.00 dB, 20 Hz −0.04 dB, **10 Hz −1.05 dB** (was −3.0), 5 Hz −18.40 dB.
- **M-12 — anti-alias stopband was ~55 dB against an ~80 dB expectation.** `scipy.signal.decimate(ftype='fir')` uses a Hamming kernel; measured **−60.0 dB** worst case, capping usable dynamic range regardless of the ADC resolution negotiated elsewhere. Replaced with a cached Kaiser design (100 dB, 0.20 transition, 259 taps at  $q=4$ ): **−111.7 dB**, passband flat to  $1e-4$  at `F_max`. The wider 0.25 transition saves 52 taps but starts eating the passband at `F_max`. The longer kernel costs nothing at block edges — measured on one block against the analytic RMS, hamming +0.2416% vs kaiser −0.0001%, because the overall's Hann taper already de-weights the edges where the start-up transient lives. Hardware A/B on identical captured samples,  $q=8$ : **+10.8 dB** at 2600 Hz and +10.3 dB at 3100 Hz where alias leakage rises above the capture noise floor; below it both kernels sit under the floor.

- **M-14 — the simulated bearing signal was not a valid oracle.** Ten pure cosines plus white noise, kurtosis ~3, no impulsiveness, no resonance carrier, no sidebands, no slip. Every diagnostic added this round keys on exactly those properties, so it could not have validated any of them: a broken envelope analyser and a correct one both return “nothing here” on ten pure cosines. Also fixed the operator-precedence bug — `int(freqs[-1] // bearing_multiple*running_rate)` binds left to right, giving ~5000× too many iterations with every out-of-range harmonic collapsing onto the last bin via `argmin`; extracted as `bearing_harmonic_count()` so the two loops needing this arithmetic cannot disagree again.
  - Two constants set by measurement, not from the textbook. **resonance\_q** : what governs impulsiveness is ring-down time over impulse period, and at the textbook  $Q=40$  the ring-down is *longer* than the gap between impulses, so kurtosis reads 3.08 against a healthy 3.09 — the fault is undetectable.  $Q=8$  gives  $\tau/\text{period}$  0.21 and separates 5.28 from 3.09. **Harmonic phase**: a single shared phase made healthy kurtosis swing 2.03–4.00 across seeds on nothing but how the cosines lined up, overlapping the faulted range; independent phases are both more physical and stable.
  - The healthy control is correctly *sub-Gaussian* (1.54–3.04), not Gaussian: a machine dominated by running-speed harmonics genuinely has kurtosis below 3.
- **S-12 — SimulatedSensor died silently above two channels.** `simulated()` ships a 2-element `scale` while the generator emits one column per enabled channel, so three channels raised a broadcast `ValueError` inside `_stream`, which had no `try/except` : the thread died while `_running` stayed set and `is_streaming` reported a healthy stream producing nothing, indefinitely. The scale is now fitted to the data, `_stream` is guarded and clears `_running` in a `finally`, and `_sample()` emits exactly one column per enabled channel (the old `max(..., N_CHANNELS)` floor gave single-channel configs a phantom second channel).
- **H-08 — AcquisitionSettings.copy()** copied only maxfreq and binsize, silently dropping every other field including all five per-channel dicts. It now round-trips through `to_dict` / `from_dict`, so a field added there cannot be forgotten here.

## Changed

- **The spectral anomaly hook is unwired from the GUI panel** (`GUI_ANOMALY_HOOK_TYPES = ('rms',)`). It fires on essentially every healthy frame: it triggers on `np.any(|spec - baseline| / baseline > threshold)` across every bin while Welch runs a single segment in every shipped preset, so each noise-floor bin is  $\chi^2(2)$  with a standard deviation equal to its own mean and  $P(\text{some bin of } \sim 2000 \text{ exceeds } 1.5\times)$  is ~1.0 — `consecutive_n` is a delay, not a defence. Bins 0 and 1 are hard-zeroed for integration, so on any velocity/displacement channel they deviate by ~1e12 and fire permanently. The hook, the config schema and the headless front end are untouched, and a stored `spectral` / `both` is preserved rather than rewritten when the dialog is merely opened. Tracked for a fix-or-remove decision as **R39**.
- Stream pacing moved out of `GenerateBearingVibration_TemporalMethod` — a signal generator should not contain a `time.sleep` — into `SimulatedSensor._stream`, which paces against a deadline so the block rate stays honest when generation is slow.
- `doc/PROGRESS.md` gains **R34–R38** (standards conformance), **R39** and **R40**. R40 records that IEPE bias monitoring is **not achievable on this hardware**: the coupler has a DC blocking capacitor so the bias never reaches the scope on either coupling setting, and the 4000A ranges only to  $\pm 20$  V against a 24 V supply, so even a direct pre-cap tap would over-range.

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## [Unreleased] — hotfix/claude-ultrareview (2026-08-29)

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Integration branch for the three-discipline audit remediation. Sections for the individual branches follow below.

## Changed

- **util.py / gui.py** — the **spectral anomaly hook is unwired from the GUI panel**.  
`GUI_ANOMALY_HOOK_TYPES` now restricts the monitor dialog's hook combo to `('rms',)`;  
`SpectralThresholdHook`, the config schema and the headless front end are untouched, so reviving it is a one-line change
  - It fires on essentially every healthy frame and has never been useful in practice. It triggers on `np.any(|spec - baseline| / baseline > threshold)` across every bin in the band, while Welch runs a single segment in every shipped preset (`nperseg == blocksize`) — so each noise-floor bin is  $\chi^2(2)$ , with a standard deviation equal to its own mean.  $P(\text{some bin of } \sim 2000 \text{ exceeds } 1.5\times)$  is  $\sim 1.0$  on healthy data, which makes `consecutive_n` a delay rather than a defence. Bins 0 and 1 are additionally hard-zeroed for integration, so on any velocity/displacement channel they deviate by  $\sim 1e12$  and fire permanently
  - A stored `spectral / both` — which headless still writes — is clamped for display only.  
`GUI._hook_type_to_save()` preserves the original, so merely opening the monitor dialog cannot silently rewrite a headless user's configuration; that is the S-07 failure mode this branch had already fixed once
  - Tracked for a fix-or-remove decision as **R39**. A real fix needs band RMS rather than per-bin, a threshold in  $\sigma$  rather than fixed %, and a default band — i.e. it depends on R34
- **doc/PROGRESS.md** — added **R39** (spectral hook fix-or-remove) and **R40** (sensor-fault detection; records that IEPE bias monitoring is impossible on this hardware — the coupler has a DC blocking capacitor so the bias never reaches the scope, and the 4000A ranges only to  $\pm 20$  V against a 24 V supply)

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## [Unreleased] — feature/peak-selection (2026-08-29)

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### Added

- **src/rev80/peaks.py** — significance-based spectral peak selection, replacing “report the top N local maxima by absolute amplitude”
  - The local noise floor is not flat: across the 60 channel-spectra of the `old castle corpus` it varies by up to  $7\times$  *within a single spectrum*. On `blower 4 - bearing DE.h5 ch2` the median local floor is 9.0 mV overall but 39.4 mV over 1890–2000 Hz, so ranking by absolute amplitude ranked by *how loud the neighbourhood is* rather than by *how far a line stands out of it* — 6 of that spectrum's top 12 came from ripple in the noisy top of the band
  - The diagnostic cost was real: 1034 and 1088 Hz are sidebands at  $\pm 25/29$  Hz around the 1059 Hz carrier — the bearing-fault signature — and ranked 10th and 4th, so at the shipped display count of 6 the sideband family was broken up and pushed off the table. The instrument was hiding the fault evidence behind noise ripple
  - `local_noise_floor()` estimates the floor per bin with a running median, then `select_peaks()` passes **array-valued** height and prominence to a single `find_peaks` call, so admission is evaluated bin by bin against each line's own neighbourhood. Everything that passes is reported; the count becomes an output and the user knob becomes a significance threshold in dB
  - **Amplitude reporting is unchanged** — the reported value is still the amplitude of the maximum bin, with no frequency interpolation and no energy summation across a peak. Ranking is still by descending amplitude: significance decides *whether* a line is reported, amplitude decides where it sits in the table
- **tests/test\_peak\_selection.py** — 626 lines, including an opt-in real-corpus regression suite (`REV80_CORPUS_DIR`, skipped by default since `DEVDATA/` is gitignored)

## Changed

- **collector.py** — `process_sample()` step 6 calls `rev80.peaks.select_peaks` instead of `find_peaks(..., distance=5)` + top-N sort. `n_segments` is derived from the Welch parameters rather than assumed to be 1, so the median-to-mean floor correction stays right if `nperseg` ever stops equalling `blocksize`
- **sample.py** — `AcquisitionSettings.peak_threshold_db` (default 9.5), persisted in `to_dict / from_dict`
- **gui.py** — the “Peak Display” count spinner is replaced by a “Peak Sig., dB” threshold. The count still exists but only as a clutter cap on the table and plot markers (default 50, against a measured corpus median of 40 and p90 of 49), and a text line reports how many lines passed and whether the cap is hiding any — without it, a capped table is indistinguishable from a spectrum that genuinely had few significant lines

## Measured

- **Default threshold 9.5 dB is set by a false-alarm cliff, not taste.** On pure noise (2001 bins, no lines at all) the mean reported count is 37.9 at 6.0 dB, **0.8 at 9.5 dB**, 0.0 at 12.0 dB
- **Floor window width 65 bins** tuned against the corpus, not argued. Scoring single-frame estimators against a 16-frame-averaged reference floor over 20 files × 3 channels × 4 frames gives a broad flat basin from 31 to 65 bins (median |error| 1.097 dB at 31, 1.039 at 45, 1.085 at 65, 1.326 at 129). A *synthetic* spectrum with a deliberately smooth floor prefers 129 — the disagreement is the point
- **Edge handling: truncated window, not the edge replication first proposed.** Replication copies one random bin 32 times, and 32 copies of a single exponential draw dominate a 65-sample median: median |error| 1.480 dB / p95 5.246 dB, against 0.591 / 1.829 dB truncated. Zero-padding (`scipy.signal.medfilt`) is worst, at −3.568 dB mean bias
- Corpus-wide at the default: count median 40, p10 22, p90 49. Under the old top-10 rule 6.67% of reported peaks stood less than 2× above their local floor (18.17% with a flat-top window); the gate admits none
- `wlen` is mandatory with `prominence` — unbounded, a single broad hump spanning the band has prominence 34.6 and its apex looks like the most prominent thing in the spectrum
- Known cost, asserted by a test so it cannot drift silently: **broad features are rejected on purpose.** A 40-bin- $\sigma$  hump’s apex has prominence 7 against a requirement of 144 and never reaches the table. That is right for a line table and wrong if you want broadband resonances flagged; they remain visible in the plot only

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## [Unreleased] — fix/measurement-validity (2026-08-28)

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Nine measurement-validity defects (F-1 ... F-9) from the vibration-engineering audit, plus one adjacent logging defect (S-09).

**Root cause of the whole class: the test suite was degenerate.** `tests/test_sample.py` sets `F_max=10000`, `binsize=2` → `fs=32768`, `N=16384`, so bin spacing is exactly 2.000 Hz — then swept tones at 500, 1000, 1500 ... 9500 Hz, **every one an exact multiple of 2**. That is the single case where the block is genuinely periodic in `N` samples, the FFT’s circular-wrap discontinuity vanishes, and the integration error is identically zero. `_make_dc` also defaulted `highpass_enabled=False`, so the filtered path was never exercised by any amplitude assertion, and the displacement tests asserted only on the spectrum peak, never on `result.overall`. Phase 1 of this branch was therefore a deliberately **red** commit: 110 tests, 92 failing, before any fix was written.

The same mistake was then made again, and caught: every tone in the new suite was generated as `sin(2 $\pi$ ft)`, so **every record started at phase 0** and `x[0]` equalled the DC level exactly — degenerate in precisely the same

way bin-centred frequencies are. Real hardware exposed it. `tone()` now takes a `phase` argument defaulting to 0.7 rad.

## Added

- `src/rev80/_dsp.py` — windowing helpers for frequency-domain integration, carrying the measured error tables that justify each choice
- `sample.py` — `AcquisitionSettings.nperseg` and `.binsize_actual`: the Welch segment length and the bin width actually delivered
- `collector.py` — `filter_block()`, `filtered_data_for()`, `reset_filter_state()`, `_seed_zi()`: persistent per-channel high-pass state
- `gui.py` — `derive_acquisition_preview()`, a single source of truth for the dialog preview
- `monitor/anomaly.py` — `valid_results()`, a shared validity filter for anomaly hooks
- `tests/test_measurement_validity.py` — 123 tests covering F-1 ... F-9 and S-09, all off-bin, with the high-pass enabled, across block boundaries and at every preset

## Changed

- `util.py` — `MAXFREQ_PRESETS` is `[2e2, 5e2, 1e3, 2e3, 5e3, 1e4]`; **the 20 kHz and 50 kHz presets are removed** and `F_max` now tops out at 10 kHz (see F-3)
- `sample.py` — `VibeSample.samplerate / ChannelResult.samplerate` / the HDF5 `samplerate` attribute are now `float`. The true rate is generally not an integer
- `sample.py` — `n_fft_bins` counts **displayed** lines (DC...`F_max`), not the full one-sided transform: 1001 at the default preset, not 2049
- `gui.py` — the spectrum info panel reports `binsize_actual`, and is labelled `F_max` rather than `AA`
- `monitor/writer.py` — no longer carries its own divergent copy of `_write_channel_group`
- CRLF → LF in `picoscope.py` and two `examples/` scripts

## Fixed

- **F-1** `collector.py` — **FFT wrap leakage corrupted every integrated overall and waveform.** `rfft` was taken on the raw, un-windowed block and multiplied by  $(j\omega)^n$ . The DFT treats the record as periodic; unless it is exactly periodic in `N` samples there is a step discontinuity at the wrap point whose spectrum is broadband and low-frequency-weighted, and `n < 0` amplifies it by  $1/\omega^{|n|}$ . The in-code claim that the round-trip was lossless held only for `n_ord == 0`, and zeroing bins 0 and 1 left the leakage in bins 2, 3, 4 ... The spectrum was never affected (Welch already windows); what was wrong was the Overall card, the trend, `overall_json`, the anomaly detector's input, and the displayed waveform
  - Two treatments, because the consumers want different things. **Scalar overalls**: Hann taper, RMS divided by the window power gain `sqrt(mean(w2))`. **Displayed waveform**: cannot be tapered — the envelope would be plainly visible on the trace — so overlap-save instead, with a Tukey `α=0.5` window killing the wrap discontinuity and only the flat middle, where the window is exactly 1.0, returned. `time_vec` is truncated to match
  - **Cost, deliberate and documented**: integrated/differentiated traces span the middle 50% of the block
  - Measured (1.0 unit 0-pk sine, `fs=32768`, `N=16384`) — velocity overall at 61.0 Hz **+12.25%** → **+0.05%**; displacement overall at 61.0 Hz **+470.59%** → **+0.18%**, at 120.7 Hz **+777.84%** → **+0.05%**, at 501.0 Hz **+4473.76%** → **+0.00%**; on-bin 500.0 Hz was **+0.00%** before and after, which is exactly why the old suite never saw any of it. Waveform peak, displacement at 501.0 Hz: **+10176.68%** → **+0.24%**

- Taper-width sweep at the worst case (61 Hz displacement):  $\alpha$  0.05  $\rightarrow$  +335.84%, 0.10  $\rightarrow$  +7.00%, 0.30  $\rightarrow$  +4.22%, **0.50  $\rightarrow$  +1.47%**. Reflection padding was tried and is far worse (+1918%) — it doubles the effective near-DC content. Differentiation ( $n=+1$ ) verified unaffected, worst +0.018%
- **F-2 `picoscope.py` — the reported sample rate was not the rate actually used.** `_start_streaming` computed `actual_raw_fs` correctly from the interval the driver writes back, then discarded it (`self._actual_samplerate = self.config.samplerate`); the comment had conflated “hide the oversampling ratio” with “hide the actual rate”. The driver quantises the sample interval to whole microseconds, so at `F_max=2000` the hardware runs 8333.25 Hz while the app labelled it 8192 — **-1.70%**, rising to -8.25% at the (now removed) 50 kHz preset
  - **Confirmed on real hardware** (PicoScope 4424A, AWG loopback on channel A), same captured samples under two labellings: a commanded 1000.00 Hz tone read 983.062 Hz (-1.694%) and now reads 1000.012 Hz (+0.001%). Mean `|error|` across four tones **1.676%  $\rightarrow$  0.034%**. Reported as a float — rounding would reintroduce a smaller version of the same error
- **F-3 `picoscope.py` — no anti-alias filtering at all above `F_max = 20 kHz`.** `antialias_decimate()` is a no-op at factor 1, and `max(1, int(min(OSR_TARGET, CEILING / samplerate)))` truncated toward zero: 1.526  $\rightarrow$  1 at 20 kHz, 0.763  $\rightarrow$  0  $\rightarrow$  clamped to 1 at 50 kHz. The 50 kHz preset additionally requested 131072 Hz raw, **31% above** the measured `STREAMING_CEILING_HZ` — the exact condition the module docstring says makes the driver silently drop most samples while still reporting `status='OKAY'`
  - A general-purpose IEPE accelerometer has a mounted resonance at 25–80 kHz with 20–30 dB of gain, so at `F_max=20 kHz` an unfiltered 50 kHz component folds to 15536 Hz — inside the displayed band, indistinguishable from real signal, and *larger* than the real signal because of the resonance gain
  - `_choose_osr()` now uses explicit `math.floor` and requires `osr >= 2`. Only `F_max  $\leq$  10 kHz` satisfies `fs*2 <= 100 kHz`, hence the preset removal above. Override by raising `STREAMING_CEILING_HZ`, which requires re-measuring the safe continuous streaming rate on the target hardware and channel count. A maxfreq outside the presets still degrades rather than failing, but now logs a WARNING naming the frequency above which content will alias
- **F-4 `collector.py` — high-pass filter state was reset to zero on every block.** `sosfilt` was called with no `zi`, reintroducing a startup transient at the head of every frame of a continuous stream. Measured over blocks 1–3 of a 200 Hz tone at a 10 Hz high-pass: acceleration waveform peak +9.41%  $\rightarrow$  -0.000%, displacement overall +19.15%  $\rightarrow$  +0.017%; with 1000 mV of residual DC offset, acceleration overall **+14321.74%  $\rightarrow$  -0.000%**
  - **Two regimes, handled distinctly, because replay is not a stream.** Live streaming filters once per frame in `receive_data`, in order, carrying state, and caches the result on the `VibeSample` since `process_sample` is called repeatedly on the same frame. Replay/browse re-processes stored frames out of order and carries no state — verified bit-identical forward and reverse to 12 significant figures. Block 0 of a stream still shows genuine settling; there is no history to carry
  - **F-4 follow-up, found by hardware:** `sosfilt_zi(sos) * x[0]` is scipy’s documented idiom and is correct when the first sample represents the baseline — true for a step, false for anything oscillatory. On four consecutive real captures the block mean was -0.06...-0.21 mV (the true DC) while `x[0]` ranged over **36...305 mV**, and under  $1/\omega^2$  that spurious step dominated: worst-block displacement error **+4297.20%** against a settled reference. Now seeded from the block mean (-2.22%). A warm-up pass — filter the block, reuse its final state as its initial state — was measured and is *worse* (+61.10%): it imposes a periodic assumption the block does not satisfy. The branch’s own synthetic tests missed this because they used phase-0 sines, reproducing the suite’s own central blind spot
- **F-5 `collector.py`, `monitor/writer.py`, `monitor/anomaly.py` — overload and degraded frames were trended, alarmed on, and stripped on save.** Five gaps forming the classic spurious-alarm

mechanism: a clipped waveform reads high with harmonic distortion, the trend records a step change that never happened, the detector fires — and on reload the record looks clean

1. The overflow bitmask was read from whichever callback completed a block, so a callback raising overflow mid-accumulation had its flag discarded. Now latched with `|=` and cleared on emit
  2. `update_trend` ran regardless of the flags. Flagged frames are now excluded from the trend, while still being displayed and still flagged
  3. `_read_frame_group` hard-coded `overflow=False` and never set `degraded`, so the flags `_write_channel_group` had faithfully stored were never read back
  4. `monitor/writer.py` had a second, divergent `_write_channel_group` that wrote no validity flags at all — the divergence was the defect; they are now one function
  5. The anomaly hooks never read either flag. `valid_results()` is applied to both event evaluation and baseline adaptation: letting a clipped frame into an EWMA baseline poisons the reference for ~33 frames just as surely as firing on it
- **F-6 `gui.py` — the acquisition dialog computed the sample rate with 2×, not 2.56×.** It duplicated the derivation and used the bare Nyquist minimum, so at the default `F_max=2000` it advertised 4.1 kS/s, 2049 lines, 1.000 s and half the true memory while the instrument ran at 8.2 kS/s, 4097 lines, 0.500 s. Now delegates to `AcquisitionSettings`; verified equal across the full preset grid
  - **F-7 `collector.py` — the PSD cache key omitted binsize and samplerate.** Switching 2 Hz → 0.5 Hz bins returned the identical cached 2049-point, 2 Hz spectrum: the user believed they had quadrupled the resolution and nothing had changed. Masked while streaming, since each new `VibeSample` starts with `psd_mv=None`, so it bit in browse/offline mode and after loading a file
  - **F-8 `sample.py` — stated line count and bin width did not match the computed spectrum.** `n_fft_bins` returned `blocksize//2 + 1` while Welch used `nfft = int(samplerate/binsize)` and `blocksize = nextpow2(samplerate/binsize) >= nfft`. **40 of 72** preset combinations were wrong; worst `F_max=200/df=20`, a real resolution of 20.480 Hz (+2.40%) and 17 lines claimed against an actual 13. Fixed with `nperseg = blocksize`; re-verified **0 of 72** mismatch. The existing amplitude suite was unaffected — at its `F_max=10000/df=2` the old and new `nperseg` were already both 16384
  - **F-9 `gui.py`, `collector.py` — the spectrum was displayed out to `fs/2`, where alias rejection is ~12 dB.** The axis, the peaks table and `find_peaks` all ran to `fs/2 = 1.28 × F_max` — the anti-alias filter's transition band, -21.8 dB at the folding frequency and effectively 0 dB at `fs/2` itself. The whole point of the `F_max = fs/2.56` convention is that the guard band is never shown. Now truncated at `F_max` before peak-finding. The info label had also called `fs/2` the “AA” frequency, which read as a specification the instrument does not meet
  - **S-09 `picoscope.py` — the ADC overflow warning inhibit was inverted, flooding the log.** `elif ch in self._overflow_warned: remove(ch)` fired precisely when a channel was *still* clipping, re-arming the warning every other callback — measured **10 warnings from 20 callbacks**. At a 1 ms poll interval that rolls every other diagnostic out of the rotating log during exactly the run being diagnosed. The clear-down loop also had to move out of `if overflow:`, since a return to zero is the only way to observe it

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## [Unreleased] — chore/config-consistency-ci (2026-08-28)

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### Added

- `.github/workflows/ci.yml` — first CI for the project. There was no `.github/` at all: 300+ tests and nothing ran them.
  - `push + pull_request`, `ubuntu-latest`, matrix over Python 3.10–3.13 (the range the new `requires-python` floor admits)

- checkout → setup-python → pip install -e ".[dev]" → ruff check src/ tests/ → pytest tests/ -q; fail-fast: false, concurrency group cancels superseded runs
- installs libx11-6: dearpygui's \_dearpygui.so links against libX11 at load time (confirmed with ldd). No GL and no X server are needed because the suite never calls create\_viewport().
- the native PicoSDK driver is deliberately **not** installed; the suite runs against SimulatedSensor
- **Verified end to end locally**, not assumed: a clean Python 3.12 venv built exactly as CI does, with the PicoSDK driver stubbed out and DISPLAY / WAYLAND\_DISPLAY unset, gives 300 passed / 10 skipped. The picosdk git+https pin resolved and built a wheel with no special handling, so --no-deps is not needed. dearpygui==2.0.0 publishes manylinux wheels for cp310–cp313 (checked against the PyPI JSON API), so the whole matrix is covered.
- **util.py** — amplitude\_scale(mode), nearest\_interval\_preset(seconds), canonical\_hook\_type(value), hook\_type\_label(value), ANOMALY\_HOOK\_LABELS, DEFAULT\_AMPLITUDE\_MODE, DEFAULT\_ANOMALY\_HOOK\_TYPE, DEFAULT\_RMS\_ALPHA, DEFAULT\_SPEC\_ALPHA, and an explicit \_\_all\_\_
- **util.py** — 600: '10 min' added to MONITOR\_INTERVAL\_PRESETS
- **README.md** — Contributing section now documents the one-time git config core.hooksPath .githubhooks install step
- **Tests** — 7 new files, 135 tests: test\_gui\_save\_config.py, test\_icons.py, test\_monitor\_pretrigger\_scaling.py, test\_sensor\_library\_integrity.py, test\_config\_contract.py, test\_anomaly\_hook\_build.py, test\_util\_small\_defects.py

## Fixed

- **.githubhooks/pre-commit** — still ran ruff check vibechecker/ and wrote vibechecker/\_version.py after the rebrand moved the package to src/rev80/. ruff exited non-zero against a nonexistent directory, so the hook blocked every commit for anyone who had installed it. Repointed at src/ tests/ and src/ rev80/\_version.py.
  - src/rev80/\_version.py regenerated: it read rc0.4-10-g11a72f4 while git describe gives rc0.5-44-g7bf4712. **Shipped builds reported a version 34 commits behind, so a field bug report could not be tied to a build.**
  - Root cause of the silent rot: the hook is not installed by default (core.hooksPath is the stock .git/hooks), now documented in the README.
- **pyproject.toml** — four defects, each breaking a clean install:
  - requires-python was >=3.8, two minor versions below the real floor. No module uses from \_\_future\_\_ import annotations, so every annotation is evaluated at import: PEP 604 dict | None in sensor.py:46 needs 3.10; dict[str, Any] in config.py:219 and argparse.BooleanOptionalAction in \_\_main\_\_.py:17 need 3.9. pip installed happily on 3.8/3.9 and the app raised TypeError on first import. Raised to >=3.10. Swept for 3.11+/3.12+ constructs (tomllib, StrEnum, ExceptionGroup, except\*, TaskGroup, typing.Self, typing.override, itertools.batched, datetime.UTC, PEP 695 generics) — none present, so 3.10 is correct, not merely safe.
  - dearpygui was an optional [gui] extra, but gui.py:6 and icons.py:3 import it at module scope and the rev80 console script reaches both via \_\_main\_\_:main. pip install -e . — exactly what CLAUDE.md instructs — produced a rev80 command that ImportError. Moved into required dependencies; [gui] kept as an empty alias.
  - pandas and matplotlib were declared runtime dependencies but imported nowhere under src/. matplotlib was already in the PyInstaller excludes, confirming it was never needed at runtime. Removed. (Both are still used by the standalone examples/TMS\_Digital\_Audio.py and scripts/advanced\_plots.py, which sit outside the installed package.)

- `numpy`, `scipy`, `h5py`, `pyyaml`, `plyer`, `pywin32` were entirely unconstrained, so a shipped installer's contents depended on what PyPI served that day. Added lower bounds (not pins) chosen as the oldest releases that support 3.10 and carry the APIs actually used.
- **pyproject.toml / CI reproducibility** — `[tool.ruff.lint]` set only `ignore`, never `select`, so ruff linted with whatever its *current default* happened to be — and that default moves between releases. On an identical tree: **ruff 0.15.10 → 0 errors, ruff 0.16.5 → 167 errors**, none from a code change. With ruff unbounded in `[dev]`, CI would have gone red on an untouched tree at the next ruff release. Fixed at both layers: `select = ["E4", "E7", "E9", "F"]` makes the rule set explicit, and `ruff>=0.15,<0.17` bounds the version. Both versions now report clean.
- **gui.py** — `_on_sb_save_config` called `h5py.File(...)` but `gui.py` never imported `h5py` at module scope; the three other users each did a function-local import and this one did not. The resulting `NameError` was caught by a broad `except Exception` and logged as "failed to patch {session\_h5}", so the session browser's **"Save Config" button was dead code** and the message pointed the user at disk permissions. Added the module-scope import, removed the three redundant local ones, and narrowed the `except` to `(OSError, KeyError)`.
- **icons.py / gui.py** — `test_gui_build` failed on any clone that had not run `scripts/build.sh`, with an opaque `SystemError: <built-in function pop_container_stack> returned a result with an exception set`. `assets/fonts/` is gitignored and populated by `build.sh`, so the font is absent on a fresh clone and on every CI runner; `icons.load()` passed the nonexistent path to `dpg.font()`, and the failure inside the context manager surfaced from `pop_container_stack` naming neither the font nor the path — taking down all of `_create_gui()`. Now checks for the file and falls back to the DPG default with a warning. *This was the pre-existing failure standing between the suite and green, and the blocker for CI.*
- **picoscope.py** — the PicoSDK driver (`libps4000a`) is a separate native install from the `picosdk` Python wrapper, and `picosdk/ps4000a.py` instantiates `Ps4000aLib()` at import time, raising `CannotFindPicoSDKError` when the driver is absent. `picoscope.py` imported it unguarded, so `import rev80.picoscope` was fatal without the driver. Reproduced by stubbing `find_library`: the **whole suite aborts** with 3 collection errors, not 3 modules' worth of skips — note `test_antialias.py` is pure DSP and was collateral damage. Also contradicted CLAUDE.md's claim of "offline development and CI without hardware". Now degrades to `PICOSDK_AVAILABLE = False`, mirroring the guard `__init__.py:9-17` already uses for this exact import; `FindPicoScope()` returns `[]` with a warning. Verified identical results with the driver present and absent.
- **monitor/controller.py:258** — read the sensor sensitivity under `sensitivity_mv_per_eu`, a key that exists nowhere in the codebase; `session.sensor_snapshot` holds `ScopeSensor.to_dict()` output, which emits `sensitivity`. The `1.0` default therefore **always** won: the mV → EU division never happened and every burst's pre-trigger trend points came out a factor of `sensitivity` too large (~10x for a 10.2 mV/g sensor) against the post-trigger points on the same continuous plot. `engineering_units` on the adjacent line used the correct key, so the unit *label* converted while the magnitude did not — worse than an obvious break. Now rehydrates through `ScopeSensor.from_dict()` once per call so the field name can only be wrong in one place; misleading comment at :246 corrected.
- **README.md:920 / CLAUDE.md:127** — documented the same wrong `sensitivity_mv_per_eu` key. Following the README raised `KeyError` in `from_dict`, and the very next README paragraph correctly said the code divides by `sensitivity` — the two lines contradicted each other. Both fixed.
- **Sensor library could be silently and permanently destroyed.** `ScopeSensorRegistry._load_user` swallowed any parse failure with `except Exception: return []`, and `_save_user` writes whatever `_load_user` returned straight back over the file. `add()` and `delete()` both follow that load-then-save

path, so **one unreadable entry erased every calibrated sensor definition, with nothing logged**. Two independent triggers reached it: the wrong documented key above, and `config._atomic_yaml_write` using `yaml.dump` with the **unsafe default Dumper** while every reader uses `yaml.safe_load` — loading a colleague's `.h5` auto-registers its sensors with no prompt and no type coercion, so a non-scalar attribute serialised as a `!!python/object/apply: tag` that `safe_load` then refused. Fixed at all three layers:

1. `config.py:159` → `yaml.safe_dump`; unrepresentable values now fail loudly at write time and the existing file survives. Also closes a latent escalation: a writer emitting object tags means any future switch to `yaml.load` becomes arbitrary code execution from a shared measurement file. *No RCE today.*
  2. `scope_sensor.py` → `from_dict` coerces every field and rejects containers and arbitrary objects, so junk cannot reach the writer.
  3. `scope_sensor_registry.py` → a bad *entry* is logged with file, index and content and skipped; a bad *file* raises, so a read failure can never become an overwrite. Read-only callers go through `all()`, which degrades to `[]` and logs, so a corrupt file cannot crash GUI construction.
- **config.py / gui.py / headless.py** — two monitor defaults the GUI could not represent and therefore silently rewrote:
    - `hook_type`: `config.py:83` seeds `'rms'` (lowercase); the GUI combo items are capitalised and both `_on_anom_config_change` and `_build_anomaly_hook` compared raw strings, while `headless.py:159` did `.lower()`. On a fresh install `'rms'` matched neither `('RMS', 'Both')` nor `('Spectral', 'Both')`: opening Config → Monitor once **hid both hook groups, built zero anomaly hooks, and silently disabled anomaly detection** behind an Enable switch that was on. Now canonical lowercase everywhere, capitalisation demoted to a display label.
    - `interval_s`: `config.py:74` seeds `600`, which was not a preset member, so the widget showed “1 h” and saving wrote `3600.0` back — **silently changing a 10-minute logging interval to hourly, discarding 5 of every 6 measurements**. Fixed at both ends: `600` is now a preset, and the fallback picks the *nearest* preset rather than a hardcoded 3600.
  - **headless.py / gui.py** — `_build_anomaly_hook` bound `period` only inside the `if hook_type in ("rms", "both")` branch but read it in the spectral branch, so a **Spectral-only config raised UnboundLocalError**. In headless this fires *after* `collector.start_stream()`, killing the process with the PicoScope still streaming and never closed — the worst outcome for an unattended run. The two copies fail differently, which is why it survived: `gui.py` reads `period` unconditionally and always raises, while `headless.py`'s `if "spec_ewma_time" in anom_cfg and period > 0` short-circuits, so it only raises when that key is present — which is exactly what the GUI writes. Hoisted in both.
  - **Reconciled the four ways the two `_build_anomaly_hook` copies had drifted**: hook-type casing (above); `warmup` default 30 (GUI) vs 10 (headless) → 10, matching `config.py`'s seed, since at 30 the GUI needed a 3x longer baseline warm-up during which nothing could fire; `spec_n` default 3 (GUI) vs 10 (headless) → 10, since at 3 the GUI fired on a third of the evidence headless required; and the EWMA-alpha fallbacks, previously hardcoded separately in each copy, now `util.DEFAULT_RMS_ALPHA / DEFAULT_SPEC_ALPHA`.
  - **\_pico\_loader.py:24** — computed `Path(__file__).parent.parent / 'drivers'` → `src/drivers`, which does not exist; `_paths.py:25` already gets this right with three `.parent`s. `ensure_pico_dlls_loadable()` silently returned `False` in development, bundled DLLs were never registered, and picosdk fell back to walking `%PATH%`. Frozen builds use the `sys._MEIPASS` branch and were unaffected. Now delegates to `_paths.resource_path('drivers')` — the duplication was the bug — and logs a warning on Windows when it returns `False`.

- **collector.py:1187,1224** — deprecated `datetime.utcnow()` replaced with `datetime.now(timezone.utc).replace(tzinfo=None)`, deliberately **naive** to match `utcnow()`'s exact semantics. See follow-ups.
- **AMPLITUDE\_SCALE.get() fallback inconsistency** — called with two different defaults across five sites: `np.sqrt(2)` in the live path (`collector.py:231, :550`) and `1.0` in the reload path (`collector.py:992, :1147, monitor/controller.py:267`). An unrecognised mode reconstructed a loaded trend **1.414x off** relative to the live trend on the same plot. All five now call `util.amplitude_scale()`; the fallback is `'0-P'` because every call site already normalises with `or '0-P'` before the lookup. Unknown modes are logged. A test asserts no `AMPLITUDE_SCALE.get()` survives anywhere in `src/`.
- **util.py** — no `__all__`, so `rev80/__init__.py`'s `from rev80.util import *` re-exported `np` and every imported name into the top-level namespace. Added an explicit `__all__`. `data_dir` is listed deliberately: it is imported into `util` from `rev80._paths` and reached as `rev80.data_dir()` by six call sites in `gui.py / headless.py`, so omitting it would have broken them at runtime rather than at import.
- **18 ruff errors** cleared (unused imports/variables, one `E401`). The five unsafe `F841`s were handled by hand: `test_vibechecker.py:186`'s `old_sr` was kept and *asserted on* — `test_load_offline_adjusts_maxfreq` captured the pre-load samplerate but never compared against it, so the “adjusts” behaviour it is named for went unverified. No `noqa` added anywhere.

## Reverted

- **.python-version** — briefly changed to `3.13` on the premise that the literal `vibecheck` was a stale string. That premise was wrong: `vibecheck` is a real `pyenv-virtualenv` holding every project dependency, and naming a `virtualenv` there is correct `pyenv-virtualenv` usage. The change resolved to a bare interpreter with no packages and produced 15 collection errors. Reverted; the `pyproject.toml` changes from the same commit stand.

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## [Unreleased] — refactor/event-pipeline (develop)

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### Changed

- **collector.py** — removed `callbacks` dict entirely; all consumers now read from `frame_cache` via `new_frame_event` rather than receiving samples directly
  - `collect_sample()` rewritten: `new_frame_event.clear() / wait() / frame_cache[-1]` instead of a `_one_shot` closure pinned into `callbacks`
  - `_data_callback()` simplified: appends to cache and sets event; no callback fan-out
- **collector.py** — `data_callback` renamed `_data_callback` (internal-only convention)
- **gui.py** — internal methods renamed with leading underscore: `display_frame` → `_display_frame`, `poll_new_frames` → `_poll_new_frames`, `create_gui` → `_create_gui`
- **gui.py** — removed vestigial `callbacks['plots']` registrations from `_on_load_file`; file-load display now goes through `new_frame_event` → `_poll_new_frames` exclusively, eliminating a latent DPG thread-safety bug (hardware-thread `_display_frame` call)
- **gui.py** — trend plot (`_update_trend_plot`) now called unconditionally in `_display_frame` so loaded HDF5 trend data is rendered in offline browse mode
- **gui.py** — removed two merge-artifact duplicate method definitions: `poll_new_frames` (stale single-quote copy) and `_on_save_click` (old DPG dialog version)

- **gui.py** — removed duplicate `ACQ_NOTES` widget block in `_create_gui` (caused DPG “alias already exists” crash on `test_gui_build`)

## Refactored

- **Tests** — `callbacks` references replaced with `frame_cache` reads across `test_vibechecker.py`, `test_multichannel.py`, `test_picoscope.py`, `test_scope_sensor.py`

## [Unreleased] — hotfix/hpf-integration-fix (2026-08-28)

### Fixed

- **collector.py** — `process_sample()` highpass + integration blow-up (field-reported spurious spike at ~1–2 Hz)
  - Root cause 1: the integration transfer function ( $(2\pi f)^n$ , applied at all three sites — 5-order overalls, PSD, time-domain output) only zeroed the exact DC bin; bin 1 (1x binsize) received the full multiplier applied to residual near-DC energy and dominated the whole spectrum (0.313 in/s at bin1 vs. a real 0.747 in/s tone peak in a captured reference file)
  - A first fix attempt weighted the transfer function by the highpass filter’s frequency response (`scipy.signal.sosfreqz`); this suppressed bin1 by ~50,000x but, being a smooth multiply across many bins rather than an exact single-bin removal, behaved as a wide kernel under circular convolution and badly distorted the time-domain signal at both block edges
  - Fixed instead by hard-zeroing bin1 directly, unconditionally (not gated on `highpass_enabled`) — zeroing an exact FFT bin removes one Fourier basis component losslessly with no boundary sensitivity, the same property that already made the bin0 zero safe
  - Root cause 2 (“edge wobble”, found while investigating #1): the zero-phase highpass (`sosfiltfilt`, introduced in feature/anti-alias) effectively doubles filter order via its forward+backward pass, overshooting the raw signal by 35–45% at both block edges for a low cutoff over a short block (10 Hz over a 1s/4096-sample block); reverted the highpass filter back to causal `sosfilt` (~8–10% startup transient)
  - Verified against real hardware captures (`DEVDATA/hpf-10hz.h5`, 78 Hz / 0.75 in/s-OP test tone): bin1 now exactly 0.0, overall amplitude 0.760 in/s vs. an expected ~0.75, residual edge softness reduced from 350%/246% to ~30–55%
- **Tests** — 4 new regression tests in `test_sample.py`: bin1 hard-zeroed for integration regardless of `highpass_enabled`, bin1 left untouched for differentiation and passthrough, and a bound on time-domain edge overshoot for the causal highpass

## [Unreleased] — feature/anti-alias (2026-08-26)

### Added

- **picoscope.py** — mandatory anti-alias oversample/decimate stage in `PicoScopeStream`
  - Field incident: a high-frequency bearing-fault harmonic aliased into the low-frequency band at low apparent power, injecting spurious spectral energy; root cause was driving the ADC directly at the target analysis rate with no anti-alias filtering
  - Always oversamples the ADC (`effective_osr`, up to 4x, capped by `STREAMING_CEILING_HZ` — a ceiling measured on real hardware: continuous `ps4000aRunStreaming / GetStreamingLatestValues` silently drops most samples above ~100–250 kHz depending on

- channel count, with no error indication), applies a zero-phase anti-alias filter, then decimates back to `config.samplerate` via the new `antialias_decimate()` helper — fully transparent to `DataCollector` and everything above it
- Electrically verified on a PicoScope 4424A with siggen loopback: a tone above target Nyquist that aliased to a spurious 125 mV peak under naive decimation is suppressed 61.9 dB by the real pipeline; an in-band tone near maxfreq passes through unattenuated
- **picoscope.py** — second, independent streaming-rate watchdog flags ( `.degraded` ) sustained USB throughput below `_RATE_DEGRADED_THRESHOLD` of the requested raw rate — the same silent data-loss mode uncovered during the anti-alias investigation, invisible to the existing silence watchdog since callbacks keep firing with `status='OKAY'` ; deliberately never triggers `_try_recover()` (a USB/bus bandwidth ceiling, not a device hang)
- **sample.py** — `VibeSample / ChannelResult` gain a `degraded: bool` field
- **tests/test\_antialias.py** — regression tests for `antialias_decimate()` : aliasing tone suppressed >20x vs. naive decimation, factor=1 no-op, legitimate low-frequency tone survives intact, independent multi-channel handling
- **tests/test\_picoscope.py** — `TestRateDegradationWatchdog` : monkeypatched-clock coverage of `_check_rate_degradation()` (healthy/degraded/recovery), confirms `_try_recover()` is never called for this condition

## Changed

- **sample.py / util.py** — `samplerate` now derives from `nextpow2(2.56 * maxfreq)` instead of 2x, guaranteeing  $\geq 28\%$  Nyquist margin for the anti-alias filter's transition band (matching the ratio commercial FFT vibration analyzers use); `MAXFREQ_PRESETS` drops the 100k/250k/500k Hz entries — hardware measurement showed these already exceed this PicoScope's continuous-streaming ceiling, silently corrupting captured data
- **collector.py** — `receive_data()` threads the `degraded` flag from each incoming frame onto every channel's `VibeSample / ChannelResult` and persists it into saved `.h5` files alongside `overflow` ; new `DataCollector.stream_degraded` property mirrors `is_streaming`
- **collector.py** — `process_sample()` 's lowpass Butterworth block removed (anti-aliasing is now mandatory upstream in `PicoScopeStream` ); highpass filter switched from causal `sosfilt` to zero-phase `sosfiltfilt` (reverted in hotfix/hpf-integration-fix, below)
- **gui.py** — Lowpass checkbox/field removed from the acquisition dialog; status line shows the auto-derived AA cutoff instead of “LP ...” text; a warning line appears when the stream is degraded

## Removed

- User-facing `lowpass_enabled / lowpass_fc` fields — they ran after the ADC had already sampled and so could never actually prevent aliasing; anti-aliasing is now mandatory and handled at capture time

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## [Unreleased] — rebrand-to-rev80 (2026-08-19 – 2026-08-28)

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From this point forward the product is named **Rev80** (formerly vibechecker). Earlier sections below retain the historical “vibechecker” name as an accurate record of the codebase at the time.

## Changed

- Rebrand vibechecker → Rev80 across the codebase, packaging, and docs
  - Package moved from `./vibechecker/` to `./src/rev80/` (src-layout; history preserved via `git mv` ); all imports updated `from vibechecker` → `from rev80`

- `pyproject.toml`: package name `rev80`, entry points `rev80 / rev80-headless`, `where=["src"]`, `pythonpath=["src"]`
- User data dir moved to `~/Documents/Rev80/`; config dir moved to `~/.config/rev80/`
- GUI title/label, CLI prog name, installer (`installer/rev80.iss`), build spec (`rev80.spec`), and icons (`assets/icons/rev80.ico / .svg`) renamed to match; README/PROGRESS docs updated
- `_paths.py` fix: `resource_path()`'s base needed one more `.parent` after the src-layout move added a directory level (it was resolving into `src/` instead of the project root); `logger.py` now resolves `logging.yaml` directly relative to its own file instead of through `resource_path()`
- **build/collect\_pico\_dlls.py** / `build.sh` — search a bundled `vendor/` folder for PicoSDK DLLs first, removing the requirement to install PicoSDK system-wide before building
- Repo root reorganized: loose docs (`CHANGELOG.md`, `PROGRESS.md`, `VibeGui Project.md`) moved into `doc/`; dev scripts (`build.sh`, `render_progress.sh` → `render_md.sh`, `runtimes.ipynb`) moved into `scripts/`; `rev80.spec` moved into `build/` alongside `collect_pico_dlls.py`, with `ROOT` updated to resolve from `SPECPATH`'s parent so spec-relative paths still point at the repo root; stale `requirements.txt` and `picosdk-install.md` removed
- `.gitignore` — corrected the PyInstaller build-output path exclusion left stale by the repo reorg
- Project management docs revisited — outstanding requirements and tech debt items reviewed and updated

## [Unreleased] — hotfix/welch\_leakage (2026-08-18)

### Fixed

- **collector.py** — Welch window functions (Hann, Blackman-Harris, etc.) introduce spectral leakage that inflates the reported overall vibration level by a window-dependent factor (Hann:  $\sqrt{3/2}$ ); overall amplitude is now derived from the RMS of an exact-inverse time-domain reconstruction instead of summing spectral peaks
  - The 5-order (`-2 ... +2`) mV RMS overalls are now computed via `irfft` of the integration-scaled `rfft` — reusing the same plain, unwindowed `rfft` already computed once for the time-domain output step, so the round-trip is lossless — as `sqrt(mean(time_ord**2))` instead of `sqrt(sum(psd_ord**2))`
  - The target-unit overall (step 7) now reuses the cached 5-order mV RMS column instead of re-deriving it from the spectrum
- **collector.py** — `process_sample()` now guards against the resolved integration order falling outside `[-2, 2]`; previously an out-of-range order silently indexed the wrong overalls column, producing false scaling — it now logs an error and returns `None` for the frame

## [Unreleased] — feature/monitor\_mode (2026-05-28 – 2026-06-30)

### Added

- **monitor/ package** — Monitor Mode interval datalogger
  - `gate.py`: `IntervalGate` with snap-to-grid scheduling and burst mode
  - `session.py`: `MonitorSession` frozen dataclass (later gains `acq_snapshot / channel_snapshot / sensor_snapshot`, `cooldown_enabled / cooldown_s`)
  - `anomaly.py`: `AnomalyHook` protocol + `NullAnomalyHook` stub
  - `writer.py`: `MonitorWriterThread` (daemon, disk-space guard)

- `controller.py`: `MonitorController` orchestrating gate/writer/anomaly
- `gui.py`: Monitor card with config tab and arm/disarm controls (relabeled and reworked repeatedly through the branch — see Changed)
- 51+ new tests across `test_monitor_gate.py`, `test_monitor_index.py`, `test_monitor_controller.py`
- Session storage iterated through several revisions in-branch:
  - v4: monitor captures write the standard metadata+frames HDF5 layout so `collector.load_data()` can open them directly with no adapter; a `capture_trigger` root attr distinguishes monitor files from manual saves
  - v5 (Phase 2 storage redesign): one `session.h5` per session (`DEVDATA/monitor/{session_id}/`) replaces per-capture files and the SQLite index entirely — `/monitor/{N}/` groups for interval captures, `/burst/{burst_id}/{frame_index}/` groups for burst events, `/burst.attrs['burst_list']` JSON for fast browser rendering without loading frame data
- **Burst capture** — `manual(trigger_burst())` and anomaly-triggered bursts with a pre-trigger ring buffer
  - Session browser rewritten as a two-tab modal (Monitor / Burst views); `resize_frame_cache()` now expands the cache before load so all captures in a session are browsable, not just the most recent `cache_frames`
  - A run of alignment fixes during the branch: the trigger frame was double-counted in the pre-trigger snapshot; `_burst_all_results` indices were misaligned against pre-trigger frames; burst `rel_time` rebased to the trigger frame (`t=0`) instead of session-relative time; frame cache resized before loading burst frames (previously silently evicted pre-trigger frames at the default 32-frame cache); pre-trigger overalls pre-computed at capture time (`_compute_pretrigger_overalls()`) rather than reprocessed at load; disk-usage estimate corrected to include the pre-buffer, not just burst duration; trend unit conversion restored when browsing a loaded session (sensor wiring was missing on the session-load path, so channels fell back to raw mV)
- **Anomaly detection hooks** (`monitor/anomaly.py`)
  - `RmsThresholdHook` — EWMA self-calibrating per-channel baseline, triggers when `|current - baseline| / baseline` exceeds a threshold for N consecutive frames; tracks streak onset time so `trigger_time / trigger_rel_time` reflect anomaly onset rather than the confirmation frame
  - `SpectralThresholdHook` — initially a stored-baseline bin-by-bin dB comparison with an optional `fmin/fmax` band; later rewritten to an EWMA per-bin baseline with a percentage threshold and peak-frequency reporting in the trigger reason
  - `CompositeAnomalyHook` — tries each hook in order, returns the first event; propagates `reset_baseline()` to all children
  - `FixedThresholdHook` — independent upper/lower level triggers with unit conversion via `UNIT_TO_SI`
  - `ewma_alpha_from_time(tau, dt)` helper lets EWMA settings be specified as a time constant  $\tau$  (seconds) instead of the opaque `alpha` value; GUI shows a live computed- $\alpha$  label
  - Post-burst cooldown gating replaces arm/disarm entirely: the anomaly hook is supplied once at `start()` and active for the whole session; a cooldown deadline set after any burst fires suppresses further triggers while interval captures continue normally
  - GUI: Anomaly Detection section in the Monitor config tab (hook-type combo, RMS/Spectral/Fixed-level/cooldown settings groups, tooltips throughout); “Reset Baseline” replaces the old Arm/Disarm button
  - 15+ new tests (`test_monitor_anomaly.py`) covering warmup gating, consecutive-N triggers, baseline set/reset, composite fallthrough, streak tracking, cooldown
- **Headless CLI** (`headless.py`) — interval datalogger without the GUI
  - `python -m vibechecker.headless / vibechecker-headless` console script: discovers a PicoScope (or `--device sim`), loads saved device config, starts the stream, runs

- `MonitorController` , prints periodic status, and shuts down cleanly on SIGINT/SIGTERM with a session summary
- `--headless` flag routes `__main__.py` into headless mode; shares the collector/monitor/sample/picoscope pipeline unchanged with the GUI path
- `--from-file` loads an `.h5` file or monitor session directory on startup (both GUI and CLI); `--[no-]autodetect` controls device discovery, defaulting off when `--from-file` is given
- `--init-config` seeds `~/.config/vibechecker/` with `acquisition.yaml` and `devices/picoscope-defaults.yaml`
- Monitor/anomaly settings configured once in the GUI persist to device config and are read back by headless as defaults, overridable by CLI args
- `dearpygui` made an optional dependency ( `pip install -e .` for headless-only installs; `pip install -e '[gui]'` for the GUI)
- **`drivers/install-picoscope4000a-driver.sh`** — Linux PicoScope driver install script; registers `/opt/picoscope/lib` with `ldconfig` after install
- **`gui.py`** — Frame info card in the right panel: timestamp, block size, and sample rate always shown; burst-browse-only fields (frame time relative to trigger, burst ID, trigger type/timestamp, max overall per channel); session ID when browsing a session or burst
- **`gui.py`** — CommitMono Nerd Font icons throughout the left panel (card headers, action buttons, browse-nav arrows); font auto-downloaded on first build and bundled into the frozen app
- **`gui.py`** — global keyboard shortcuts: Ctrl+A autoscale, Ctrl+K start/stop, Ctrl+S save, Ctrl+O load, Ctrl+Q quit, ←/→ frame browse (live only)
- **`gui.py`** — configurable frame cache depth ( `AcquisitionSettings.cache_frames` , default 32) with derived recording-window and memory-usage display

## Changed

- **`config.py`** — layout split into `acquisition.yaml` (maxfreq/binsize/monitor/anomaly, per instance), `devices/picoscope-<model>-<SN>.yaml` (channels + siggen, per device), and `devices/picoscope-defaults.yaml` (new-device template); `device_config_path()` now takes ( `model_name` , `serial_number` ) and produces human-readable filenames; no migration path — delete `~/.config/vibechecker` to reseed
- **`collector.py`** — overalls now always computed over the full FFT spectrum; the `trend_fmin / trend_fmax` “Trend Frequency Window” band-limiting knob removed entirely (dataclass field, config default, UI tags, dialog widgets, tests)
- **`collector.py`** — `nperseg` clamped to signal length in the Welch PSD call to avoid spurious warnings on short blocks
- **`collector.py`** — `_load_v3 / _load_v4` consolidated into a single `load_data()` method, fixing a v4 load crash where the old `_load_v4` delegated to `_load_v3` , which read a `data` key that doesn't exist on v4 trend groups
- Version tooling: git-describe version written by a `post-commit` hook, later moved to `pre-commit` so `_version.py` is included in the commit that changes it; installer version now derived from `_version.py` at build time instead of hardcoded in the `.iss` file
- Build fixes accumulated through the branch: UPX disabled (was corrupting the frozen `python3XX.dll` 's PE import table); `pandas` removed (pulled in a `pytz` version-detection failure in frozen builds; the peaks table now uses `list[tuple]` ); `plyer` 's Windows filechooser dependency ( `win32com / pywintypes` ) added to hidden imports; `pyyaml` hidden-import name corrected ( `yaml` , not `pyyaml` ); `dist/` cleaned and any running instance killed before rebuilding; `assets/` bundled into the frozen app (font was missing, causing a startup crash); PyInstaller cache wipe no longer runs by default ( `./build.sh all clean` to force it); Inno Setup arch identifier updated to `x64compatible`

## Removed

- `monitor/index.py` (SQLite session index) — superseded by the single- `session.h5` v5 layout
- Arm/disarm as a user-facing concept — anomaly detection now runs for the whole monitored session, gated only by post-burst cooldown

## [Unreleased] — refactor/mv-domain-trend (2026-05-05)

### Changed

- **sample.py / collector.py** — `VibeSample` now stores raw mV throughout; all sensitivity conversion, Butterworth filtering, and Welch PSD computation moved into `DataCollector.process_sample()`
  - `VibeSample`: `process()`, `_convert_time_domain()`, `push_sample()`, `save()`, `load()` removed; adds `overflow: bool` and cached `psd_mv / freq_hz / _psd_config_key / overall_ampl_by_integration_order (5)` mV RMS fields
  - `ChannelResult` gains an `overflow` field
  - `receive_data()` becomes a pure mV pass-through (no sensitivity or filtering applied)
  - `process_sample(ch, sample): filter → Welch PSD (cached) → 5-order overalls (cached) → sensitivity + SI + integration → ChannelResult`
  - `process_samples()` becomes the collector-owned frame dispatcher; appends to trend only while streaming, uses a `-1` cursor index in browse mode
  - Trend storage changed from a dict-of-lists to `dict[int, {rel_times: ndarray, orders: ndarray(M,5)}]`; new `get_trend_for_display()` centralizes unit/sensitivity/amplitude-mode conversion so `gui.py` no longer imports `util` for it
  - `get_active_eu()` returns `'mV'` immediately when no `ScopeSensor` is assigned, avoiding nonsensical unit conversion before sensitivity is known
  - HDF5 v4: per-channel `(M,5)` orders matrix plus per-channel `rel_times`; v3 files are promoted to v4 trend structure on load (order-0 only)
- **gui.py** — display loop simplified around `process_samples() / get_trend_for_display()`; `_compute_channel_result()` removed; overflow now read from `result.overflow` instead of a bitmask
- **picoscope.py** — `_setup_siggen()` now also called from `start()`; previously it only fired on the reconnect path in `_try_recover()`, so the signal generator never started on the initial `stream.start()` call
- **collector.py** — overalls now computed over the full FFT spectrum unconditionally (groundwork later formalized by removing `trend_fmin / trend_fmax` in `feature/monitor_mode`); HDF5 load paths consolidated

### Added

- **gui.py** — global keyboard shortcuts (Ctrl+A/K/S/O/Q, arrow-key frame browse) — later extended in `feature/monitor_mode`
- **gui.py** — configurable frame cache depth with recording-window/memory display, `DEFAULT_CACHE_FRAMES` sourced from `config._BUILTIN_DEFAULTS` — later extended in `feature/monitor_mode`
- Build tooling: git-describe version written by a post-commit hook and embedded into the installer via `_version.py`; PyInstaller/Inno Setup fixes for UPX corruption, a `vibechecker.spec` merge conflict, `pandas / plyer` bundling, and `pyyaml` hidden-import naming

## [Unreleased] — feature/windows-build (2026-03-29 – 2026-05-27)

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### Added

- **\_paths.py** — cross-platform path sanitization (Phase 1): `sys._MEIPASS`-aware `data_dir()`, `log_dir()`, `resource_path()`; replaces the third-party `path` library with `stdlib` `pathlib` across all modules; `SAVEDIR / DataCollector.datadir` route to `~/Documents/vibechecker/data/` in frozen builds
- **\_pico\_loader.py** / `build/collect_pico_dlls.py` — PicoScope driver bundling (Phase 2): a Windows script locates `ps4000a.dll / picoipp.dll` from the PicoSDK install (registry + default paths), validates the 64-bit PE header, and copies them to `drivers/`; `_pico_loader.py` registers that directory via `os.add_dll_directory()` before `picosdk` import, handling both frozen (`sys._MEIPASS/drivers/`) and dev layouts
- **vibechecker.spec** / `installer/vibechecker.iss` / `build.bat` / `build.sh` — PyInstaller + Inno Setup build pipeline (Phase 3): one-dir PyInstaller build bundling `logging.yaml`, driver DLLs, and dearpygui data; Inno Setup 6 script for a 64-bit, non-admin install to `%LOCALAPPDATA%` with a soft PicoSDK-present check and Start Menu/desktop shortcuts
- App icon: placeholder icon wired into both the PyInstaller spec and Inno Setup script (real artwork deferred)
- **README.md** — restructured for users, contributors, and Windows builders: Windows install section (PicoSDK + installer), cross-platform source install section, Contributing section (dev env, project layout, icon swap guide), full prerequisites table (Python, Git for Windows, PicoSDK, Inno Setup)

### Fixed

- PicoSDK 11.x detection by DLL path when the registry key is absent
- `logging.yaml` moved into `vibechecker/` so `resource_path()` resolves it correctly in both dev and frozen builds
- Removed `scipy` submodule excludes from the PyInstaller spec — `scipy.signal` depends on `scipy.linalg` internally and the build broke without it
- Debug `print()` statements stripped from `gui.py / __main__.py`; Inno Setup branding (publisher, copyright, license) updated
- Inno Setup Compiler (ISCC) lookup corrected to the proper install `APPDATA` directory
- Icon assets moved to `assets/icons/`

### Changed

- **picoscope.py** — `FindPicoScope` replaced with multi-device enumeration via `ps4000aEnumerateUnits`, opening each device by serial number; `PicoScopeStream` gains `_open_unit_by_serial` so it targets the correct device when multiple scopes are connected
  - **picoscope.py** — hand-maintained `_channels_for_model` lookup table replaced by `_probe_channel_count`, which calls `ps4000aSetChannel` for channels A–H and counts successes, so any future hardware variant self-reports its channel count without a code change
  - Ruff lint fixes across `picoscope.py`, `sample.py`, `util.py`
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## [Unreleased] — feature/channel-naming (2026-04-03 – 2026-04-13)

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### Added

- **sample.py** — AcquisitionSettings gains channel\_names and channel\_target\_units dicts with name\_for(ch) / target\_unit\_for(ch) helpers, persisted in to\_dict / from\_dict ; later extended with channel\_amplitude\_modes , channel\_couplings , channel\_voltage\_ranges dicts and matching typed accessors ( amplitude\_mode\_for , coupling\_for , voltage\_range\_for )
- **gui.py** — offline post-analysis mode: load\_data auto-configures enabled\_channels and maxfreq from file contents; plot series, axes, and results panel sync after load; connection summary shows “File Loaded” (yellow) with frame/channel count; trend plot gets a vertical cursor line for the browsed frame position; acquisition buttons disabled when no device is connected
- **gui.py** — native file dialogs via `plyer.filechooser` (kdialog on KDE, native Win32) replace the DPG file dialog; `SAVEDIR` resolved to an absolute path so the dialog opens in the right place; browse-waveform controls consolidated into a single `_on_browse` dispatcher
- **gui.py** — Channels tab redesign: per-channel collapsing header row (color swatch + name text, then coupling/range/sensor on a second line), themed colored-when-enabled / grey-when-disabled; enabled state moved outside the collapsing header with a target-unit/amplitude indicator on the header itself; amplitude-mode combo added per channel; color indicators switched from a `■` glyph (didn't render in the app font) to `drawlist / draw_rectangle`
- **util.py** — expanded `UI_Elements` tag registry: named constants replace previously hardcoded `DEVSETUP_*` / `SREG_FIELD_*` strings; new tags for the redesigned channel rows ( `scope_ch_name_text` , `scope_ch_hdr_theme` , `scope_ch_amplitude_mode` )
- File Handling card gains a multiline Measurement Notes widget, read on save and populated on load
- Tooling: `ruff` added to dev dependencies with a `.githubhooks/pre-commit` hook ( `git config core.hooksPath .githubhooks` to activate); `pyproject.toml` sets line-length 120, ignores E402/E701

### Changed

- **collector.py** — HDF5 format migrated v1 → v2 → v3 across the branch
  - v2: metadata moved into `.attrs` ( /acquisition , per-frame/per-channel groups) instead of child datasets; `_load_v1` retained for back-compat
  - v3: structured /metadata/ group — /metadata/acquisition (scalars only), /metadata/scope\_sensors/{id} (sensor library, each unique sensor stored once), /metadata/channels/{ch} (name/unit/coupling/voltage\_range/sensor\_id/target\_unit/amplitude\_mode); /frames/{i}/{ch}/data stores raw samples only; /trend/rel\_times becomes a shared axis with /trend/{ch}/data per channel; `load_data` dispatches to `_load_v3` only, `_load_v1` / `_load_v2` removed; `save_data` overwrites existing files instead of early-returning
  - Fixed a file-mode bug where the channel list in the config dialog was derived from `enabled_channels` rather than the frame cache, so disabling a channel made it disappear from the dialog entirely
- **scope\_sensor.py** — `amplitude_mode` field removed; it's a per-channel acquisition setting, not a sensor property, and moves to `AcquisitionSettings.channel_amplitude_modes` (old files with the key are silently ignored on load)
- **gui.py** — config dialog scroll fix: outer window gets `no_scrollbar / no_scroll_with_mouse` , the tab bar and each tab's content are wrapped in their own `child_window` so scrolling stays contained per-tab, and the Close button is pinned outside the tab area
- **sample.py** — spectral peak detection: display count default raised (3 → 6) and relabeled “Peak Display”; minimum peak distance changed from a spectrum-length-relative value to a fixed 5 bins

## [Unreleased] — refactor/queue-handoff (2026-04-08)

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### Changed

- **collector.py / gui.py** — decouple collector → GUI with `threading.Event`, precursor to the fuller refactor/event-pipeline cleanup above
    - `DataCollector.new_frame_event`: set by `data_callback` and `reprocess_last_block`
    - `GUI.poll_new_frames`: polls the event each render tick and grabs `frame_cache[-1]`
    - Switched from `dpg.start_dearpygui()` to a manual render loop
    - Removed the `callbacks['plots']` registration — the GUI no longer hooks directly into the collector for display; `collect_sample`'s one-shot capture still used the `callbacks` dict at this point (removed entirely later, in refactor/event-pipeline)
    - Naturally handles GUI lag: rendering always shows the latest frame and skips intermediates, while all frames remain in `frame_cache` for browsing regardless of render rate
  - **gui.py** — guarded `poll_new_frames` against a rare `IndexError` race if the hardware thread shifts the deque between `len()` and indexing; silently skips the frame since fresh data arrives next tick
  - **README.md / CLAUDE.md** — architecture docs updated for the event-based collector↔GUI decoupling: callback diagrams replaced with the event-signal + poll-loop pattern; stale `sounddevice` references removed
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## [Unreleased] — feature/picoscope

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### Added

- **vibechecker/picoscope.py** — PicoScope 4000A acquisition backend (Phase 1)
  - `FindPicoScope()` — enumerates connected PS4000A units; returns `VibeSensor`-compatible dicts with `unit=['mV']`, mirroring the `FindDigiducer` interface
  - `PicoScopeStream` — background polling thread wrapping `ps4000aRunStreaming`
    - Converts ADC counts → mV via `adc2mV` on every driver callback
    - Accumulates variable-sized chunks into exact `blocksize` blocks before firing `DataCollector.receive_data`
    - Implements `.active / .start() / .stop() / .close()` interface (compatible with `sounddevice.InputStream` and `SimulatedSensor`)
    - Handles USB-only / non-USB3 power states (status codes 282 / 286)
    - Reads back actual achieved sample rate after `ps4000aRunStreaming` and updates `AcquisitionSettings.samplerate`
- **AcquisitionSettings** — two new PicoScope-specific fields (`sample.py`):
  - `voltage_range: int = 8` — PS4000A range index (8 = PS4000A\_5V)
  - `coupling: str = 'AC'` — channel A input coupling ( 'AC' or 'DC' )
- **util.py** — extended `SAMPLERATES` list to include PicoScope-relevant rates: 100 kHz, 200 kHz, 500 kHz, 1 MHz
- **util.py** — added 'mV' to `SUPPORTED_UNITS / UNITS` dict for raw voltage passthrough
- **examples/ps4000a\_triangle\_stream\_plot.py** — standalone script: generates a 500 Hz triangle wave (0.5 V amplitude, +1.4 V DC offset) via the PS4000A signal generator, streams Channel A at 50 kHz for 100 ms, then renders a Plotly HTML report with Welch PSD (10 windows, 50 % overlap) and top-5 peak detection

## Changed

- **sensor.py** — `VibeSensor.find()` now calls `FindPicoScope()` instead of `FindDigiducer()`; `VibeSensor.connect()` returns a `PicoScopeStream` for hardware sensors and a `SimulatedSensor` for the simulation path
- **sensor.py** — removed `sounddevice` import and the `sounddevice` reset workaround; renamed internal `_callback` → `_sd_callback` (simulation path only)
- **collector.py** — removed `sounddevice` import; broadened `PortAudioError` catch in `start_stream()` to `Exception`; rewrote `recieve_data()` channel extraction to handle both `(N, channels)` 2-D arrays and 1-D arrays, with channel index clamping
- **sample.py** — `VibeSample.get_accel()` wraps `convert_units` in a try/except so unsupported conversions (e.g. `'mV' → 'g'` before sensitivity is applied) pass through raw data instead of raising
- **README.md** — added PicoScope Integration section: architecture change, new `AcquisitionSettings` fields, Phase 1 data flow diagram, Phase 2 roadmap

## Removed

- `digiducer.py` / `sounddevice` no longer used in the main acquisition path (file retained for reference; `FindDigiducer` still exported from `__init__.py`)

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## [0.1.0] — main (2025-xx-xx)

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Initial public snapshot of the **sounddevice** / **Digiducer** acquisition path with:

- `VibeSensor` / `SimulatedSensor` / `DataCollector` pipeline
- `VibeSample` with Welch FFT, velocity spectrum, HDF5 save/load
- `AcquisitionSettings` with enforced interdependencies
- `dearpygui` GUI with real-time time-domain and frequency-domain plots
- Butterworth highpass filter (4th-order SOS, default 10 Hz cutoff)
- Simulated bearing-defect signals ( `GenerateBearingVibration_SpectralMethod` , `GenerateBearingVibration_TemporalMethod` )
- Comprehensive README and pytest suite